

3.2 Model-based decomposition

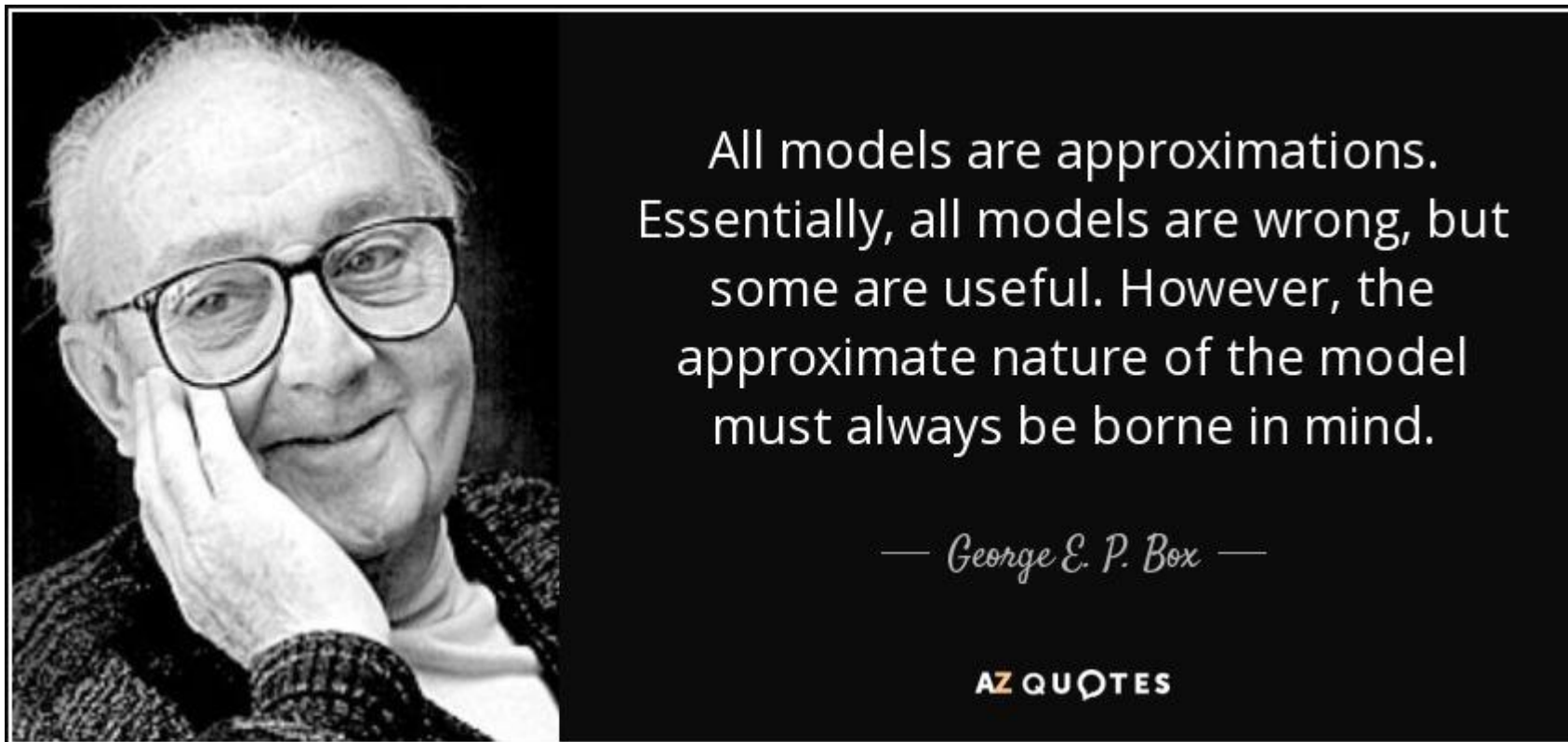
ESTP Training
05-07 October 2026

eurostat 

0. Outline

- Unobserved component models
 - STS, AMB
- SEATS algorithm
- Estimation
 - MMSE estimators, algorithms
- Analysis
 - Finite / infinite approaches
- SEATS in JD+
 - Parameters, non decomposable models, estimation

Always keep in mind...



0.1 Some notations

- Operators

- Backward, forward operator: $B^a y_t = y_{t-a}, F^a y_t = y_{t+a}$
- Differencing: $\nabla_f y_t = y_t - y_{t-f} = (1 - B^f) y_t$
- Annual sum: $S_f y_t = y_t + \dots + y_{t-f+1}$
- $\nabla_f = \nabla_1 S_f$

- ARMA

- $\Phi(B) y_t = \Theta(B) \varepsilon_t$

- ARIMA

- $\Delta(B) \Phi(B) y_t = \Theta(B) \varepsilon_t$



0.2 Useful tools

- Wold representation of an ARMA model

$$\Phi(B)y_t = \Theta(B)\varepsilon_t \Leftrightarrow y_t = \Psi(B)\varepsilon_t, \quad \Psi(B) = \frac{\Theta(B)}{\Phi(B)}$$

- Auto-covariance generating function (acgf)

$$acgf(y_t) = \sigma^2 \Psi(z^{-1})\Psi(z) = \sigma^2 \frac{\Theta(z^{-1})\Theta(z)}{\Phi(z^{-1})\Phi(z)}$$

- Example

- $y_t = (1 + .8B)\varepsilon_t \Leftrightarrow \Psi(B) = 1 + .8B$

- $acgf(y_t) = (1 + .8z^{-1})(1 + .8z) = 1.64 + .8z^{-1} + .8z$

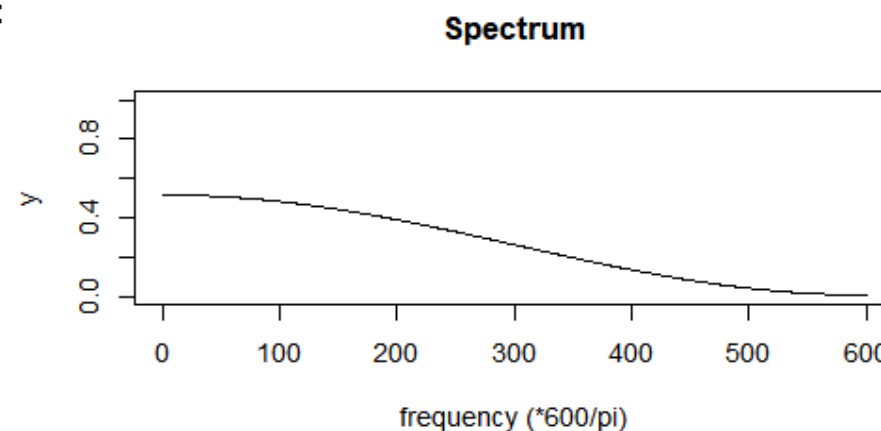
0.2 Useful tools

- Spectrum $\equiv$ Fourier transform of the acgf

$$g_y(\lambda) = \frac{\sigma^2 |\Theta(e^{i\lambda})|^2}{2\pi |\Phi(e^{i\lambda})|^2}$$

- Example (cont.)

$$\begin{aligned} g_y(\lambda) &= \frac{1}{2\pi} |1 + .8 \cos \lambda + .8 i \sin \lambda|^2 = \\ &= \frac{1}{2\pi} ((1 + .8 \cos \lambda)^2 + (.8 \sin \lambda)^2) \\ &= \frac{1}{2\pi} (1.64 + 1.6 \cos \lambda) \end{aligned}$$

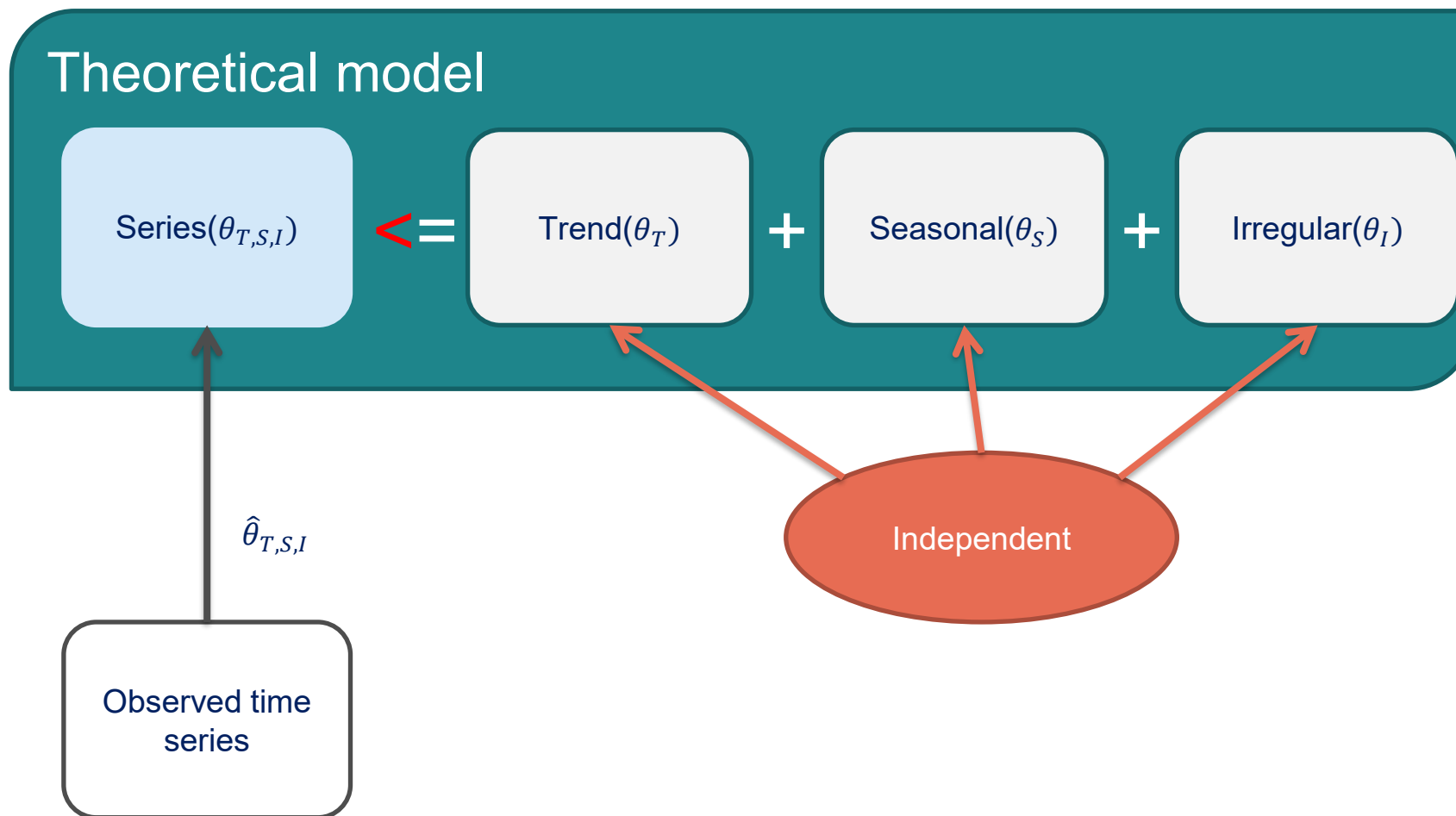


- Formal extension to the non-stationary cases

1. Unobserved components model

- Principle
 - The different components of the series are “defined” by [ARIMA] models
 - The components are independent
 - The sum of the model of the components is the model of the series

1.1 Unobserved components model (STS)



1.1.1 Basic structural model (STS)

- Local linear trend (l_t)

$$\Delta l_t = m_t + \mu_t, \quad \mu_t \sim N(0, \sigma_\mu^2)$$

$$\Delta m_t = v_t, \quad v_t \sim N(0, \sigma_v^2)$$

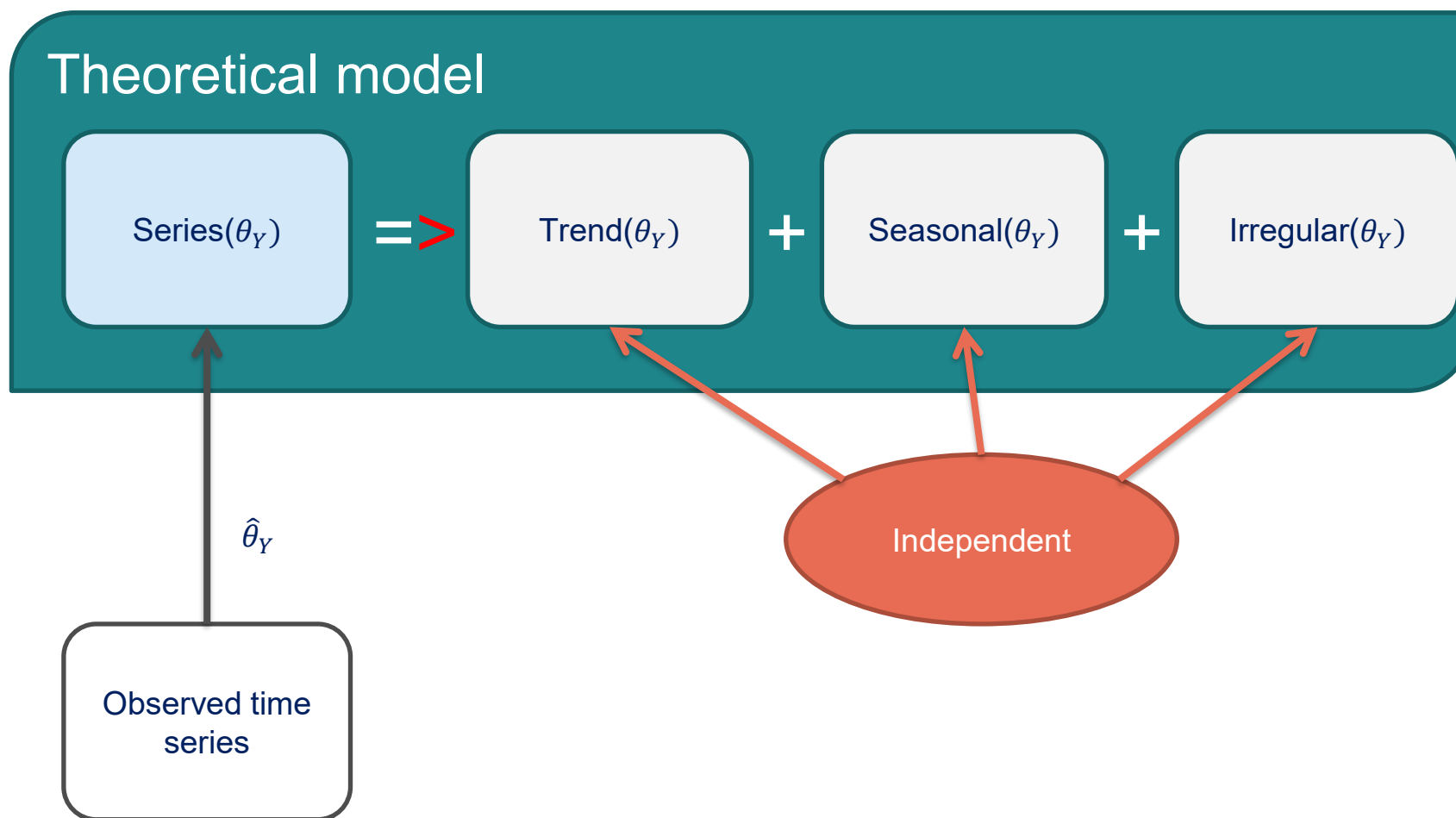
- Seasonal component (i_t)(*dummy model*)

$$S_f s_t = \gamma_t, \quad \gamma_t \sim N(0, \sigma_\gamma^2)$$

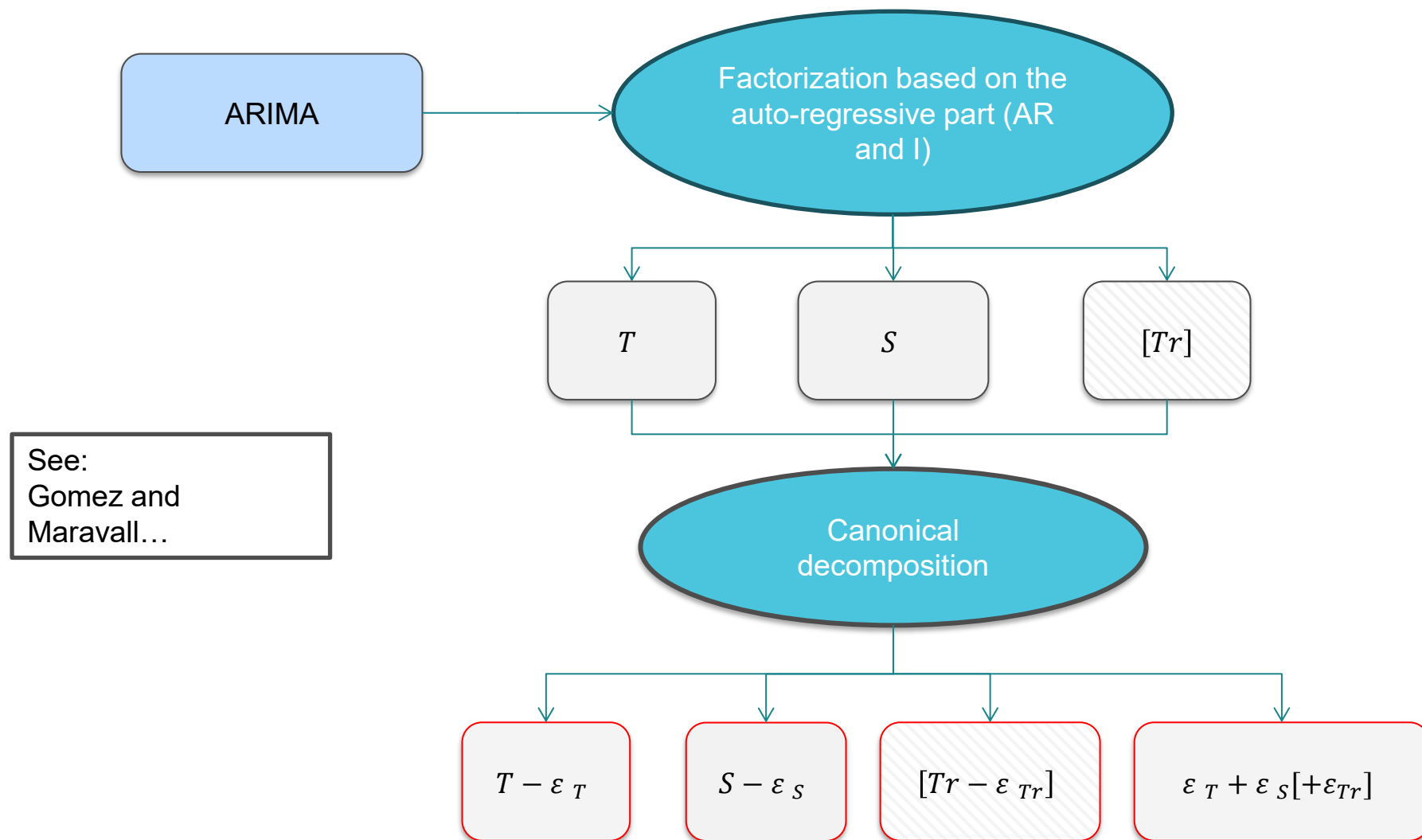
- Noise (i_t)

$$i_t = \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2)$$

1.2 Unobserved components model (AMB)



2.1 SEATS algorithm



2.2 SEATS Algorithm. Details (I)

- ARIMA:

$$\Phi(B)\Delta(B)y_t = \Theta(B)\varepsilon_t$$

- Factorization of the AR polynomial:

$$\Phi(B)\Delta(B) = \prod (1 - \alpha_i B)$$

- Trend-cycle

- α_i real,
 - $\alpha_i \geq k$
- α_i complex,
 - $|\alpha_i| \geq k, \arg(\alpha_i) \leq c$

$$\left(c = \frac{\pi}{s}\right) \sim \text{cycle length} \geq \text{two years}$$

TrendCycleSelector.java

```
public boolean accept(final Complex root) {
    Complex iroot = root.inv();
    if (root.getIm() == 0) {
        return iroot.getRe() >= m_bound;
    } else {
        if (iroot.abs() >= m_bound) {
            double arg = Math.abs(iroot.arg());
            if (arg <= m_lfreq) {
                return true;
            }
        }
        return false;
    }
}
```

- Seasonal

- α_i real,
 - $\alpha_i < -l$
 - α_i complex,
 - $|\arg(\alpha_i) - f_s| \leq e$

f_s seasonal frequency

SeasonalSelector.java

```
public boolean accept(final Complex root) {
    if (Math.abs(root.getIm()) < 1e-6) {
        if (1/root.getRe() < -m_k)
            return true;
        else
            return false;
    }

    double pi = 2 * Math.PI / m_freq;
    double arg = Math.abs(root.arg());
    double eps = m_epsphi / 180 * Math.PI;
    for (int i = 1; i <= m_freq / 2; ++i) {
        if (Math.abs(pi * i - arg) <= eps)
            return true;
    }
    return false;
}
```

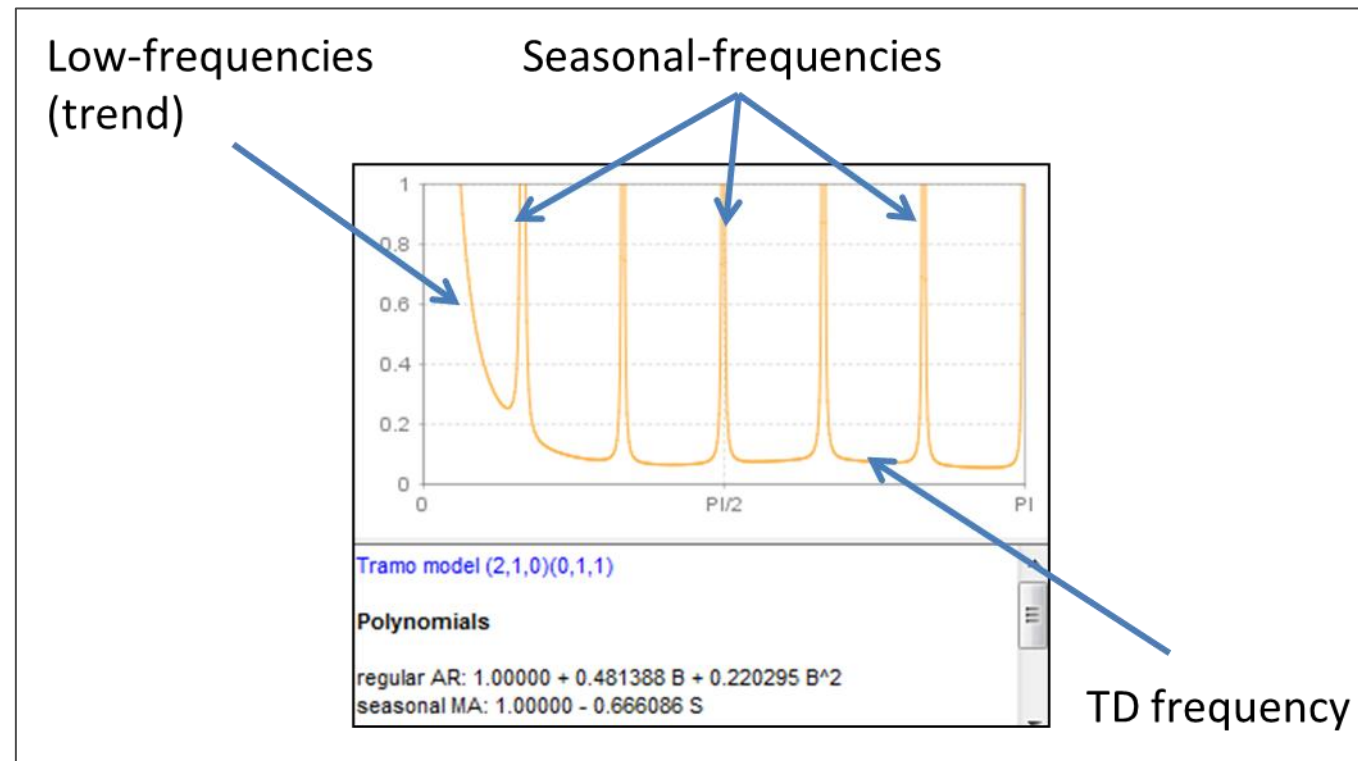
- Transitory (I)

- All other roots

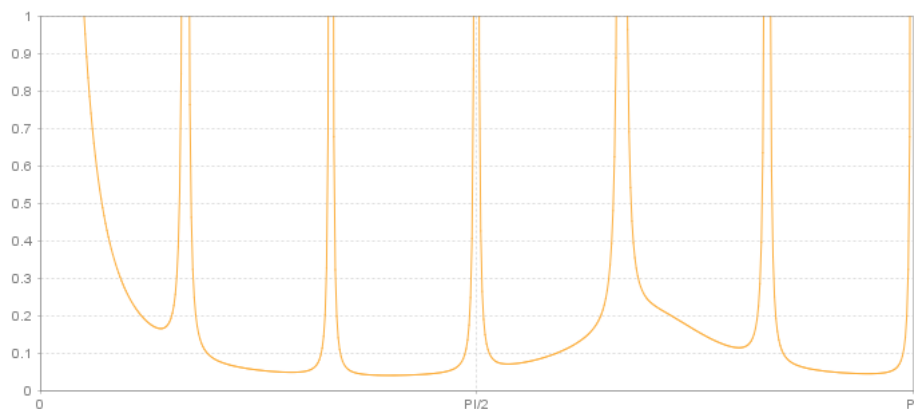
Trend boundary	k
Seasonal tolerance (degree)	e
Seasonal boundary	l
Seas.boundary (unique)	l (no seasonal part)

2.3 SEATS = spectral decomposition

- Splitting of the spectrum following the different « types » of frequencies



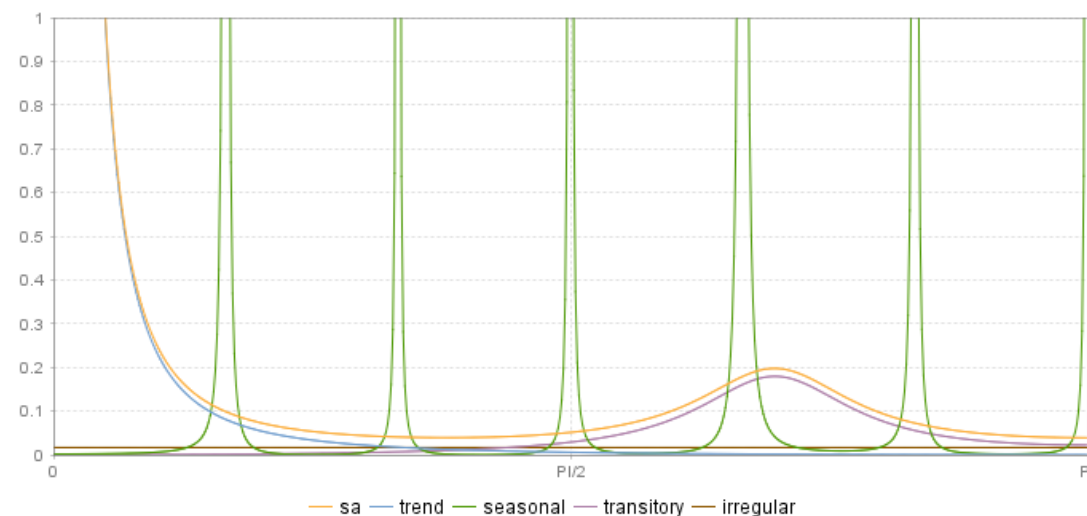
2.3 SEATS = spectral decomposition



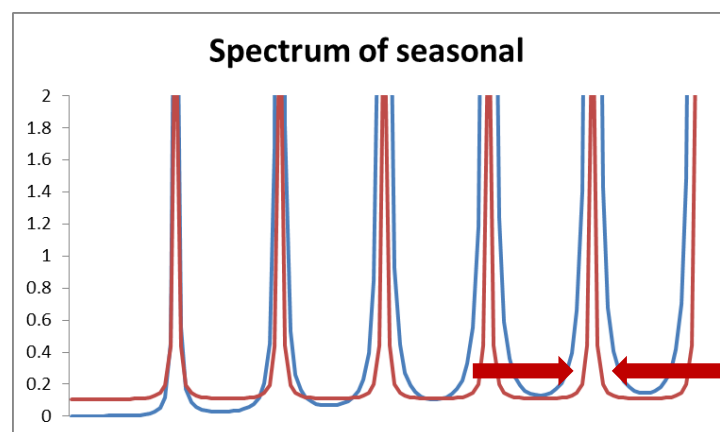
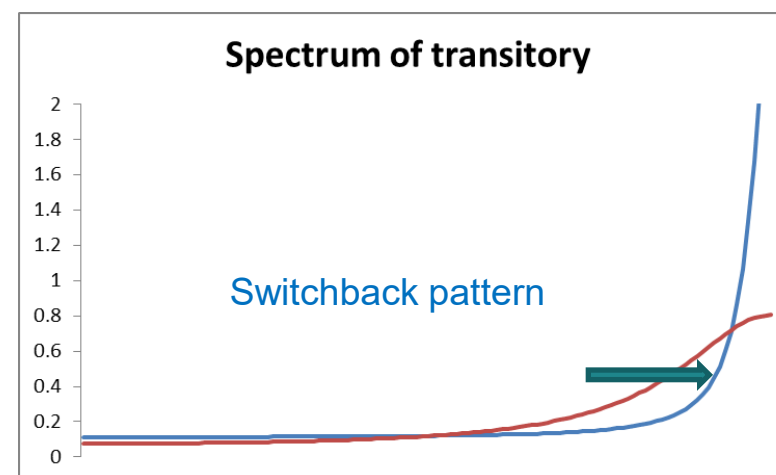
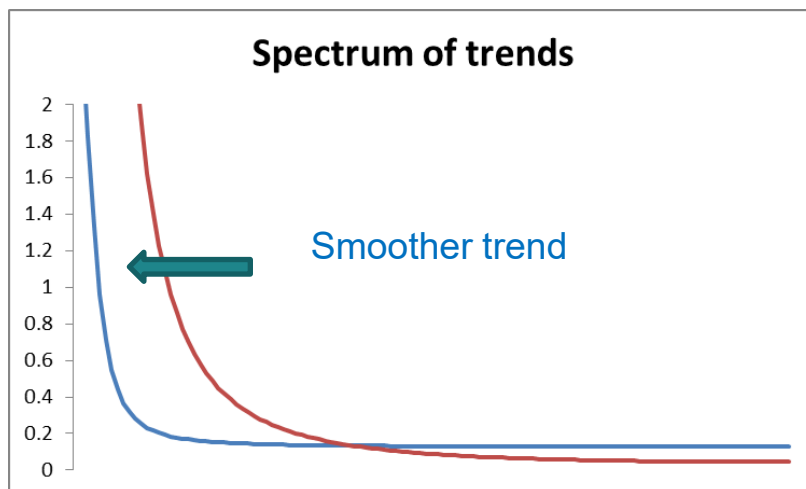
$$S_Y = S_T + S_S + S_{Tr} + S_I$$

$$S_{SA} = S_T + S_{Tr} + S_I = S_Y - S_S$$

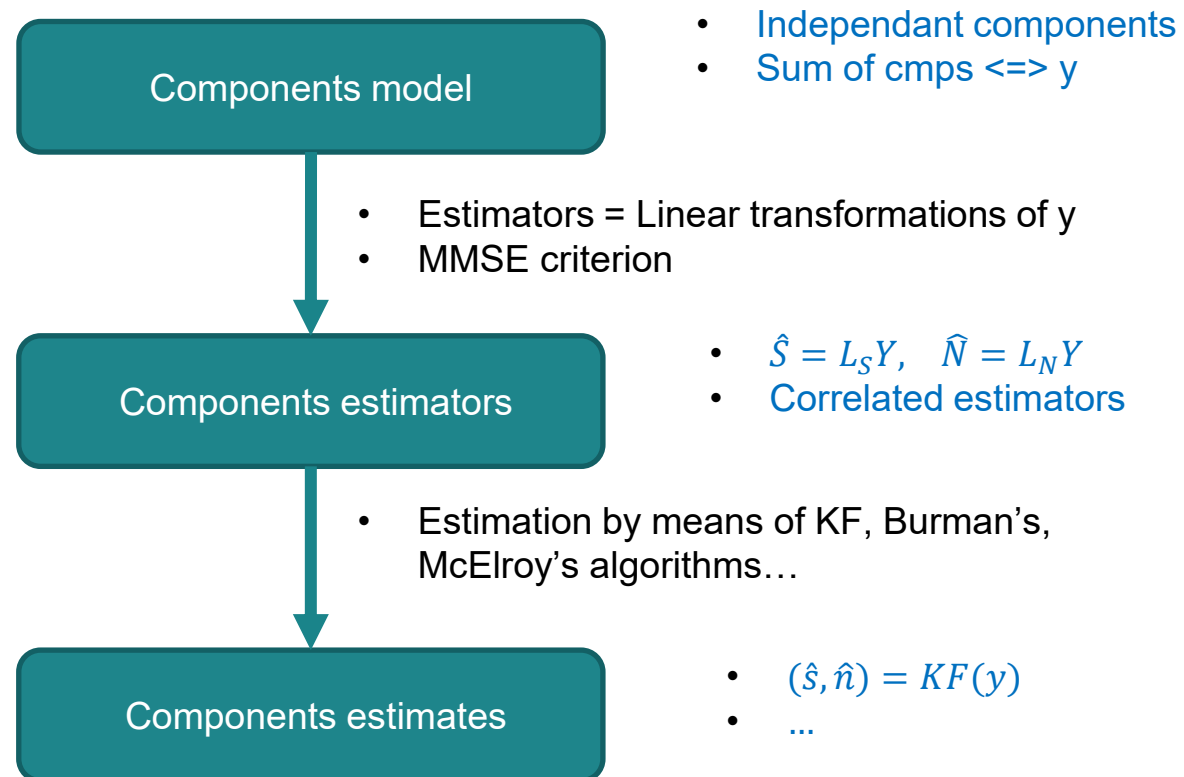
Independent components !



2.3 SEATS = spectral decomposition



3. Estimation of the model



3.1 Estimation of the components

- STS and AMB → UCARIMA models
 - Same estimation algorithms
 - Common analysis tools
- 3 solutions, strictly equivalent (except for SD)
 - Burman algorithm (WK filters): legacy and default solution, fastest, rich analysis framework, (semi-)infinite approach
 - Kalman smoother: more stable but complex, exact finite properties
 - [Matrix computation]: Not available in JD+ 3.x



3.1.1 Components

- $\Phi(B)y_t = \Theta(B)\varepsilon_t$
- $\Phi_s(B)s_t = \Theta_s(B)\varepsilon_{s,t}$
- $\Phi_n(B)n_t = \Theta_n(B)\varepsilon_{n,t}$

- $y_t = n_t + s_t$
 $\Downarrow$
- $\Phi_y(B) = \Phi_s(B)\Phi_n(B)$
- $\Theta_y(B)\varepsilon_t = \Phi_n(B)\Theta_s(B)\varepsilon_{s,t} + \Phi_s(B)\Theta_n(B)\varepsilon_{n,t}$

3.1.2 Estimators

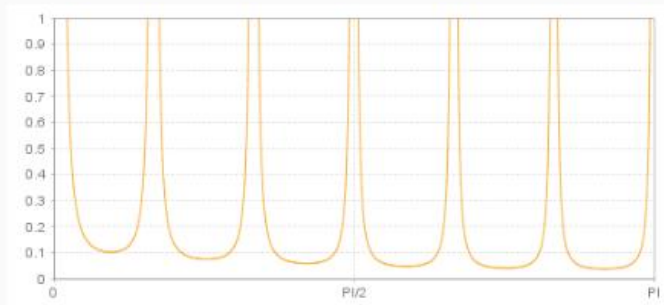
MMSE criterion



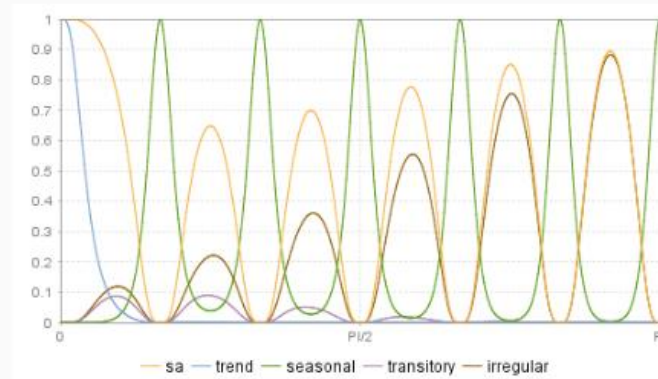
Wiener-Kolmogorov filter (Whittle, 1963)

- $\hat{s}_t = k_s \left[\frac{\Theta_s(B)\Phi_n(B)\Theta_s(F)\Phi_n(F)}{\Theta(B)\Theta(F)} \right] y_t$
- $\hat{s}_t = k_s \left[\frac{\Theta_s(B)\Phi_n(B)\Theta_s(F)\Phi_n(F)}{\Theta(B)\Theta(F)} \right] \left[\frac{\Theta(B)}{\Phi(B)} \right] \varepsilon_t = k_s \left[\frac{\Theta_s(B)\Theta_s(F)\Phi_n(F)}{\Phi_s(B)\Theta(F)} \right] \varepsilon_t$
- $\hat{s}_t = \sum_{i \leq t} \xi_{s,i}^- \varepsilon_i + \sum_{i > t} \xi_{s,i}^+ \varepsilon_i$
- $\hat{s}_{t|t} = \sum_{i \leq t} \xi_{s,i}^- \varepsilon_i$ (concurrent estimator) , $r_t = \sum_{i > t} \xi_{s,i}^+ \varepsilon_i$ (revisions)

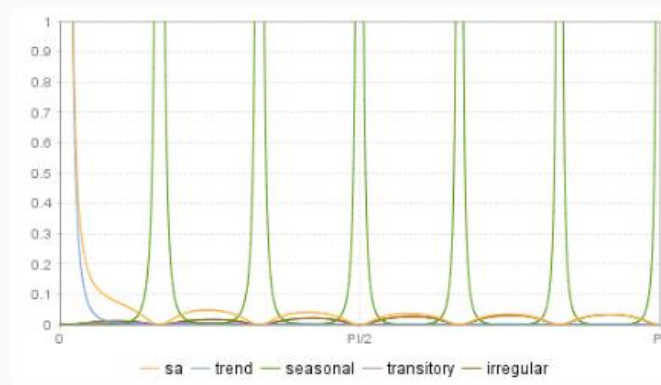
4.1 Analysis : Graphical representations



(a) $g(\omega)$



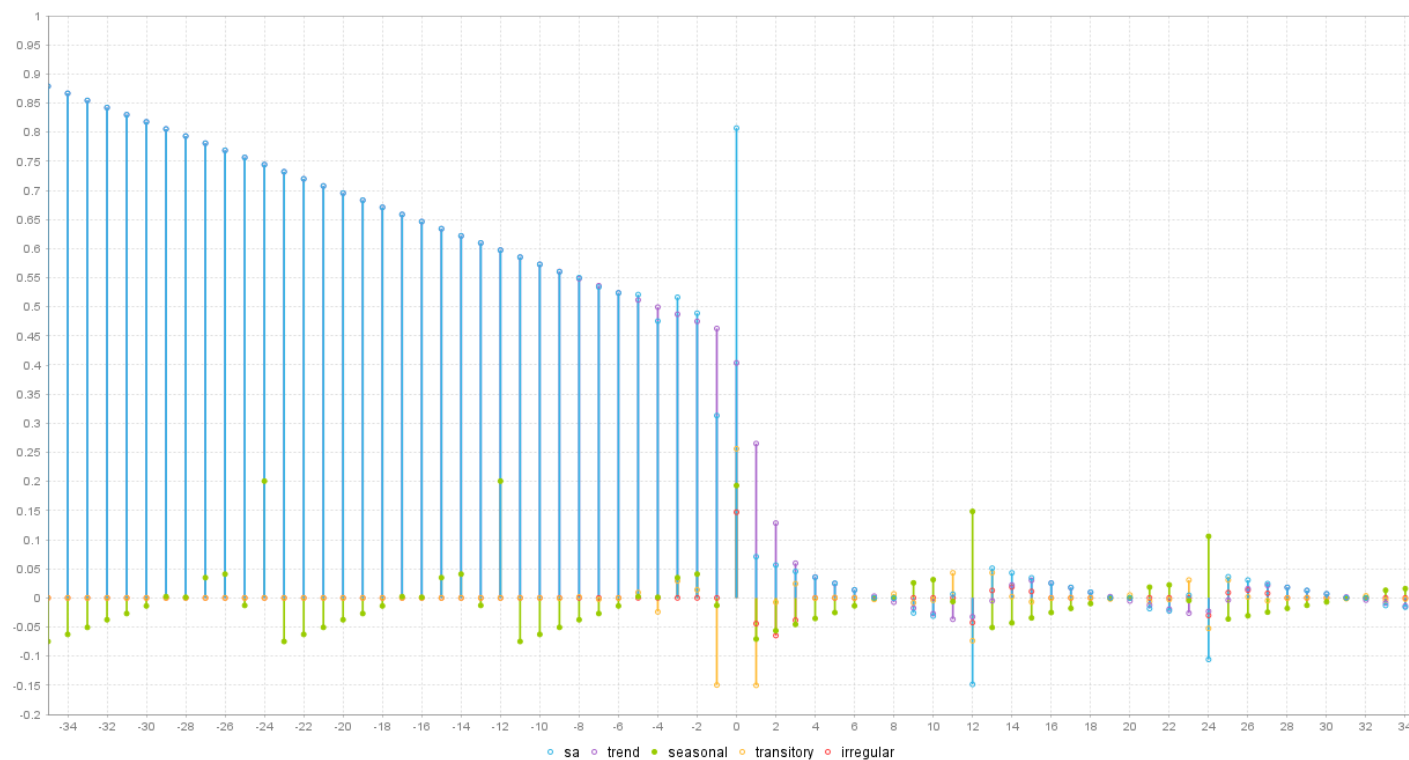
(b) $\nu_X^2(\omega)$



(c) $g_{\hat{X}}(\omega)$

4.1 PsiE-weights

- =Psi weights for estimators



4.1 Properties of the components, estimators, estimates

Distribution of component, theoretical estimator and empirical estimate (stationary transformation)

Variance

	Component	Estimator	Estimate	P-Value
Trend	0.0060	0.0001	0.0001	0.3219
Seasonally adjusted	1.4296	0.8100	0.7529	0.6307
Seasonal	0.6917	0.3057	0.2453	0.3731
Irregular	0.2967	0.1449	0.1308	0.3682

Autocorrelation

Trend

Lag	Component	Estimator	Estimate	P-Value
1	0.0036	0.6011	0.6216	0.6546
2	-0.4964	0.0708	0.1252	0.5681
3	0.0000	-0.0153	0.0821	0.3281
4	0.0000	0.0296	0.1949	0.0986
5	0.0000	0.0288	0.1484	0.2000
6	0.0000	0.0278	-0.0362	0.3585
7	0.0000	0.0267	-0.1486	0.0589
8	0.0000	0.0255	-0.1141	0.1582
9	0.0000	0.0415	-0.0615	0.2927
10	0.0000	0.0076	-0.1131	0.2189
11	0.0000	-0.1956	-0.2827	0.3205
12	0.0000	-0.3488	-0.3778	0.6880

Seasonally adjusted

Lag	Component	Estimator	Estimate	P-Value
1	-0.6263	-0.6421	-0.6869	0.3063
2	0.1103	0.1339	0.2599	0.1876
3	0.0126	0.0059	-0.1429	0.1521
4	0.0027	0.0018	0.0672	0.5279
5	0.0006	0.0004	0.0732	0.4357
6	0.0001	0.0001	-0.1026	0.1504
7	0.0000	-0.0001	0.0578	0.5349
8	0.0000	-0.0007	-0.0867	0.4062
9	0.0000	-0.0022	0.1425	0.1630
10	0.0000	-0.0510	-0.1872	0.1818
11	0.0000	0.2446	0.3076	0.4713
12	0.0000	-0.3809	-0.4096	0.6872

4.2 Analysis

- What matters?
 - Understanding the differences between the “theoretical components” and their “estimators”
 - For instance: “dips” in the spectrum of the estimator
 - Understanding the properties of the estimators
 - Model of Irregular $\neq$ white noise, negative $ac(1)$ in many SA estimators, ...
 - Understanding PsiE-weights
 - Revisions of the components = forecasts errors * PsiE weights
 - Differences between the “final” estimator and the “concurrent” estimator
- Diagnostics: comparison of the properties of the estimators and of their estimates (variance, auto-correlations...)

5. SEATS in JD+

The screenshot shows the SEATS configuration window in JD+ software. The window is divided into several sections. The 'SEATS' section is expanded, showing a table of parameters. The 'Approximation mode' is set to 'Legacy' and 'MA unit root boundary' is 0.95. The 'Trend boundary' is 0.5, 'Seasonal tolerance' is 2, 'Seasonal boundary' is 0.8, and 'Seas. boundary (unique)' is 0.8. The 'Method' dropdown is set to 'Burman'. Below this, the 'BENCHMARKING' section is also expanded, showing 'Is enabled' and 'Target' options, with 'Burman' selected for both. Three callout boxes with arrows point to specific parts of the window: 'Modification of the model' points to the 'Approximation mode' and 'MA unit root boundary' row; 'Decomposition of the model' points to the 'Trend boundary', 'Seasonal tolerance', 'Seasonal boundary', and 'Seas. boundary (unique)' row; and 'Estimation of the components' points to the 'Method' dropdown menu.

SEATS	
Approximation mode	Legacy
MA unit root boundary	0.95
Trend boundary	0.5
Seasonal tolerance	2
Seasonal boundary	0.8
Seas. boundary (unique)	0.8
Method	Burman

BENCHMARKING	
Is enabled	Burman
Target	McElroyMatrix

Annotations:

- Modification of the model (points to Approximation mode and MA unit root boundary)
- Decomposition of the model (points to Trend boundary, Seasonal tolerance, Seasonal boundary, and Seas. boundary (unique))
- Estimation of the components (points to Method)

5.1 SEATS in JD+ (cont.)

- Impact of the parameters
 - **Only in case of AR polynomials**
 - k
 - Small: possible « noisy » trend
 - $k \approx 1$: more stable trend
 - e
 - Large (>5): possible short term cycle in the seasonal (for instance, stochastic TD)→erratic seasonal
 - l
 - Small ($<.8$): higher risk of erratic seasonal
- General consideration: threshold effects are unavoidable



5.2 Non decomposable models

- Some models are not decomposable (often due to complex models)
 - Best solution: change yourself the model ($\rightarrow$ airline)
 - Otherwise:
 - Legacy: Seats search for another SARIMA model, as similar as possible to the original model
 - Noisy: Seats add noise in the initial model ($\rightarrow I = Tr[= 0]$)

6. Final remarks

- What matters?
 - Impact of the model on
 - SA/S “smoothness” $\Rightarrow$ Be very careful with stationary AR roots
 - Revisions
 - Use PsiE-weights to understand/anticipate revisions
 - Model-based diagnostics
 - Variance estimators $\leftrightarrow$ variance estimates
 - Should not happen if the original model is well defined
 - Be careful with non decomposable models / fixed models / “bad” models
 - To go further, see:
 - “SEASONAL ADJUSTMENT AND SIGNAL EXTRACTION IN ECONOMIC TIME SERIES”, by R. Gomez and A. Maravall

6. Final remarks (cont.)

- Advantages and drawbacks of model-based decomposition
 - +
 - « Optimal filters », tailored for each series
 - Many useful additional information (uncertainty...)
 - -
 - Might generate huge revisions
- Some recommendations
 - Fix the model
 - Fix the SA in the past (5 years revisions)
 - Fix the parameters and revise them when you revise the raw data
 - Be pragmatic !



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